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Main problem

One of the biggest operational issues for PT. MPM, a regional producer of product packaging based in Indonesia, was the lack of a suitable system in measuring and monitoring the performance of key business processes. In the absence of a centralized performance monitoring system, the company's senior management were not able to effectively monitor and evaluate timely accurate sales and production. The performance data was distributed along with reports from many different departments which were not connected to one another, making it very difficult for decision makers to aggregate and make sense of the information. This fragmented information process resulted in slow and defective decision making, which had a direct impact on the organization's ability to respond rapidly to customer needs, production efficiency, and the market trends.There was no single, unified Key Performance Indicator (KPI) monitoring dashboard, which resulted in the fact that important KPIs like Overall Equipment Effectiveness (OEE), Sales forecast accuracy, Production capacity utilization or Revenue performance were not monitored and analyzed in a single interface.

To counteract this systemic issue, the company did not have a competitive position relative to industry counterparts that had sufficiently good data to enable strategic decisions and it did not have real-time visibility into the performance of their operations. The need of this problem spurred the research team to create a Business Intelligence (BI) dashboard solution, developed with the waterfall sdlc methodology and Google Data Studio, to consolidate performance information and provide accurate, interactive, and real-time visualizations for both sales and production KPIs, allowing for efficient management.

Proposed Solution

To address the lack of a centralized and integrated performance monitoring system, a BI dashboard solution is proposed. The primary objective of this solution is to merge fragmented data from multiple departments into a single platform. The system is developed using the Waterfall methodology, starting from requirement analysis through system design, implementation, testing, and deployment. This methodology is suitable as the system requirements are clearly defined based on the identified operational issues.

One of the core components of the solution is the integration of data from various departments, including sales, production, and inventory. Previously, these data sources operated independently, leading to inconsistencies and duplication. To tackle this issue, Extract, Transform, Load (ETL) process is implemented.The data is collected from different operational systems and reports. Then, the data undergoes data transformation where it is cleaned, standardized, and formatted to ensure consistency. Next, the processed data is stored in a centralized data warehouse. This centralized repository ensures that all stakeholders access a single, consistent version of the data, eliminating discrepancies and improving data reliability. The integrated data is then visualized through an interactive dashboard developed using Google Data Studio. The dashboard provides real-time insights into KPIs, allowing management to monitor business performance efficiently. The dashboard consists of two main modules which are sales performance dashboard and production performance dashboard. Unlike the previous fragmented reporting approach, the proposed system provides near real-time data updates in a single dashboard interface.

User Story

1. As a Top Manager, I want to view the Sales and Production Performance Dashboard so that I can monitor organizational KPIs and make strategic business decisions quickly and accurately.

2. As a Sales Staff, I want to input and update sales data into the system so that sales performance can be analyzed and compared against forecast targets.

3. As a Production Manager, I want to monitor production KPIs such as OEE and actual output so that production efficiency can be improved.

4. As a Data Analyst, I want to conduct ETL processes and to keep the data warehouse up to date, accurate and consistent with the data sources in the operational environment, so that the information on the dashboard is accurate, current and consistent with the data sources in the operational environment.

ERD diagram using UML notation

The entities in the ERD for the BI dashboard system of PT are the main data elements needed to enable the creation of both the sales performance dashboard and the production performance dashboard. The entities that are included are customer, product, product_category, production, machine, sales, and inventory with each of the entities having its own primary key. Sales and Production are the two main fact entities in the data model, holding transactional and operational data directly used for the KPI visualizations on both dashboards.

There are many product records stored in the product_category and many product records can be used in many sales, production and inventory records as foreign keys. The relationship between customer and sales is one to many, where one customer can make many sales. In the same way, the relationship between production and machine is many to many, allowing to track the production performed by each specific machine for multiple production periods. All these entities and relationships combine to form the data warehouse that provides the feed to the BI dashboard. The design allows for all of the required dimensions of time, product, customer and machine to be sliced and filtered in the dashboard for meaningful performance insights. The foreign key relationships between the entities enable the data to be integrated from several operational sources into a single view of data that can be queried and subsequently used to create visualizations of sales and production KPIs that are accessible to management.

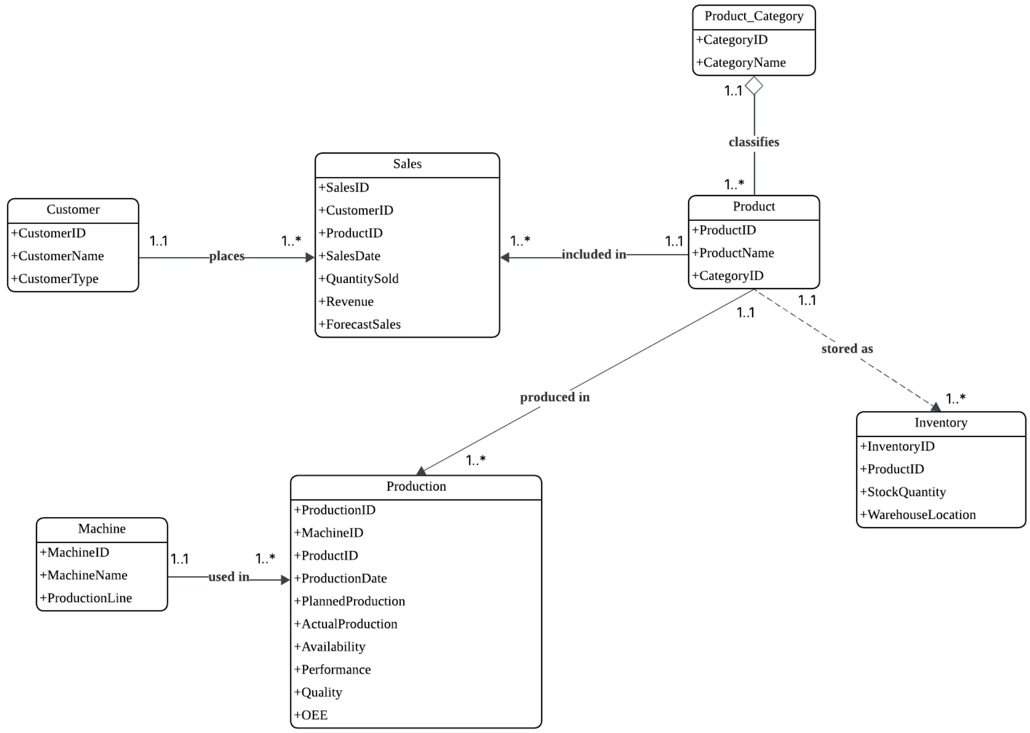

Figure 1 : ERD Diagram

Use case diagram

The BI dashboard system for PT is composed of 4 actors, each playing a different role within the organization that is involved in a specific interaction with the BI system. Top management is the main consumer of the dashboard with access to five use cases that span the entire range of dashboard use for performance monitoring, from viewing the sales dashboard to the production dashboard, to analyzing key performance indicators and comparing forecast with actual values, to producing reports. The use cases collectively support senior management to make well-informed and timely strategic decisions based on centralized and accurate data. Operational actors who put information into the system include Sales Staff and Production Manager, who respectively feed the information into the system and update the information about sales transactions. The Production Manager who inputs production information and monitors OEE and the production capacity and evaluates the efficiency of the shop floor.

The Data Analyst then is in charge of carrying out the ETL process, where in every operational department of the company, data is extracted from raw data, converted to structured and clean data, and uploaded to the data warehouse application that feeds the dashboard. They are also responsible for the ongoing maintenance and updating of the data warehouse, ensuring that the information contained in each of the dashboards is current, consistent, and reliable.

Figure 2 : Use Case Diagram

Data Architecture

Figure 3 shows the data architecture to transform raw operational data into meaningful business insights. The architecture integrates data from multiple sources, processes it through ETL operations, stores it in a centralized data warehouse, and presents the information through interactive dashboards for decision-making.

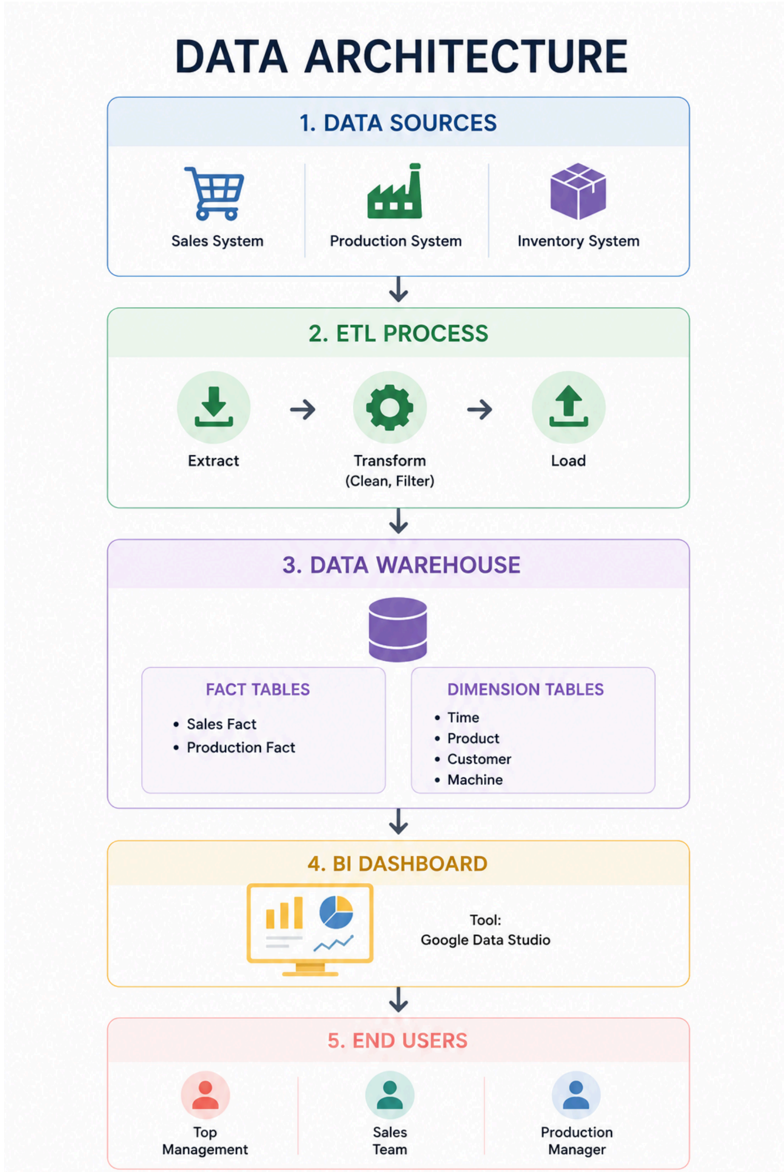

Figure 3 : Data Architecture

The Data Source Layer consists of operational systems that generate data from sales system, production system and inventory system. The sales system stores sales transactions, customer information, and product data. The production system records production output, machine performance, and OEE metrics whereas the inventory system manages stock levels and inventory movement data. These systems serve as the primary sources of information for the BI platform. The ETL process transforms the integrated data into a standardized format. This phase extracts raw data from operational systems. Then, data transformation occurs where the data is cleansed, filtered, and standardised. Finally, the processed data is loaded into the data warehouse. Thirdly, the data warehouse acts as a centralized repository for storing integrated data. The warehouse adopts a star schema structure consisting of fact tables and dimension tables. This structure enables efficient querying and reporting. Then, the processed data is visualized through dashboards developed using Google Data Studio (Looker Studio). The dashboard provides KPI monitoring, interactive charts and reports and performance tracking and trend analysis. These data visualizations allow users to interpret business data quickly and effectively. Lastly, the dashboard is accessed by various stakeholders, including top management, sales team and production manager. Top Management uses the dashboard for strategic decision-making, the sales team uses it for monitoring sales performance and customer trends, whereas the production manager uses it for tracking production efficiency and operational performance.

Multidimensional Diagram

Based on the business requirements and dashboard objectives, two separate star schemas were developed which are the Sales Performance Star Schema and Production Performance Star Schema. The Sales Performance Star Schema supports the analysis of sales activities, revenue generation, and forecasting, while the Production Performance Star Schema focuses on manufacturing efficiency and Overall Equipment Effectiveness (OEE) monitoring.

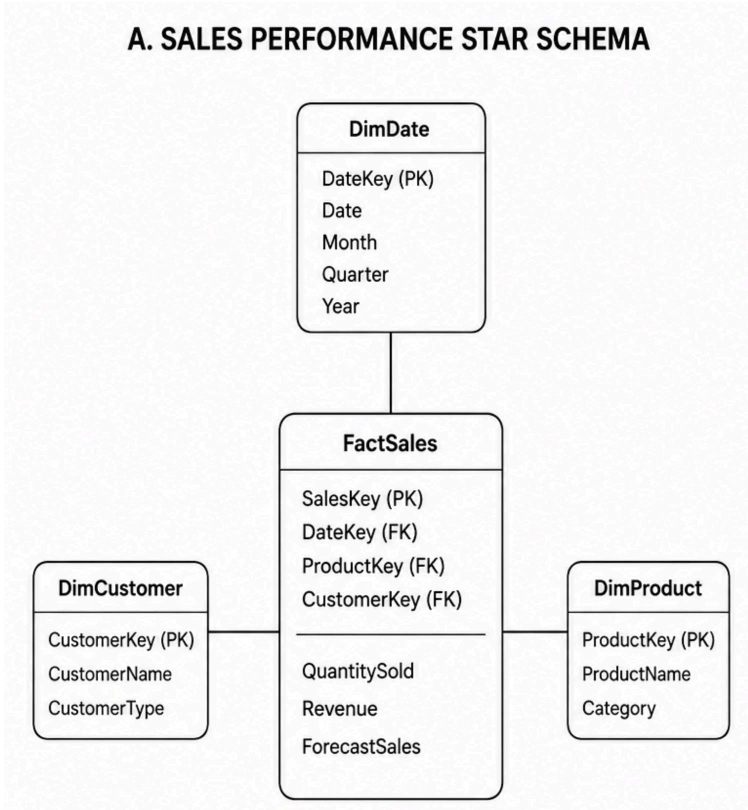

Figure 4 : Sales Performance Star Schema

The Sales Performance Star Schema is designed to support sales-related analysis and reporting. The schema consists of one fact table, FactSales, and three dimension tables: DimDate, DimCustomer, and DimProduct. The FactSales table is the central table that stores quantitative sales measurements to analyze sales performance, identify trends, and evaluate forecasting accuracy. The DimDate table provides temporal information that allows users to perform time-based analysis such as monthly sales trends, quarterly performance analysis, and annual revenue comparisons. The DimCustomer table stores customer-related information used for segmentation and customer analysis, to identify purchasing patterns among different customer groups and evaluate customer contribution to revenue generation. The DimProduct table contains product-related information used for product performance analysis, such as allowing users to evaluate product demand, compare category performance, and identify top-selling products.

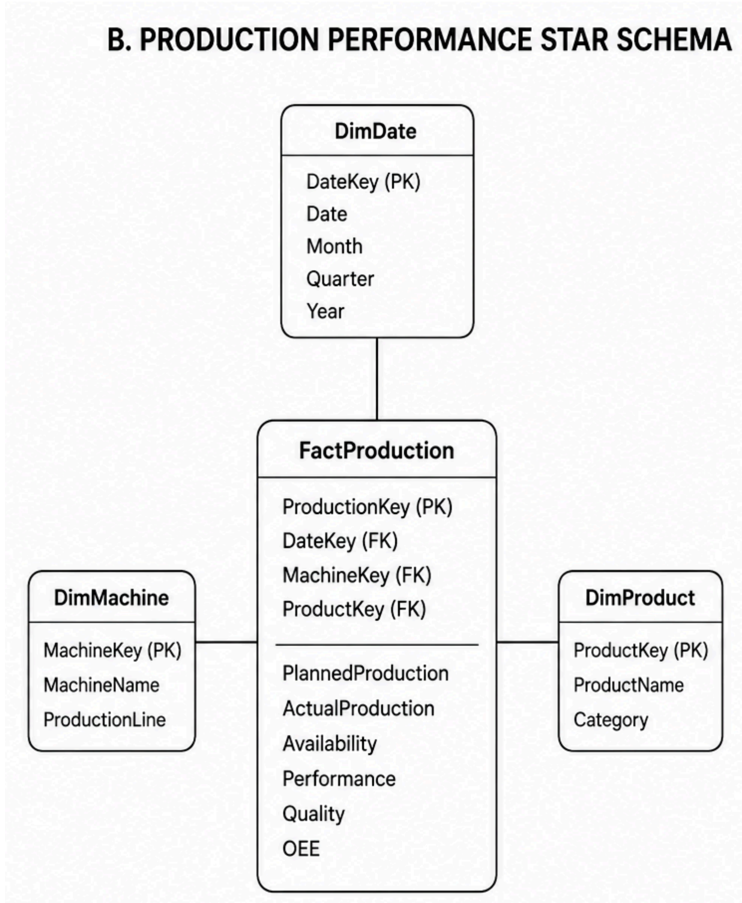

Figure 5 : Production Performance Star Schema

The Production Performance Star Schema is developed to support manufacturing performance monitoring and operational analysis. These measures enable the organization to evaluate production efficiency and identify operational issues. The DimDate table provides time-related information for production analysis which allows production managers to evaluate manufacturing performance over different periods. The DimMachine table contains information about production equipment and manufacturing lines which supports machine-level performance monitoring and production line analysis. Lastly, the DimProduct table stores information regarding the products being manufactured which enables production analysis based on product categories and product-specific manufacturing performance.

The schemas are separated because they represent different business processes. However, both schemas can coexist within the same data warehouse. Shared dimensions such as DimDate and DimProduct can be reused across multiple fact tables through a conformed dimension approach.

This design enables cross-functional analysis, allowing management to compare sales performance with production output using common dimensions such as time and product. As a result, decision-makers can obtain a more comprehensive view of business operations and improve strategic planning.